Failure Mode and Effects Analysis (FMEA)

Duration: 1 Day

Mode: Online / Classroom / On-Site

Certification: QLSS Certificate of Completion

Course Introduction

Failure Mode and Effects Analysis (FMEA) is a structured, proactive methodology used to identify potential failure modes in product design or manufacturing processes and assess their impact on performance and safety.

This training is based on the AIAG-VDA FMEA Handbook, which standardizes risk assessment approaches across automotive OEMs and suppliers. The session walks through the 7-step method for both Design FMEA (DFMEA) and Process FMEA (PFMEA), emphasizing practical application, cross-functional collaboration, and linkage with APQP and Control Plan.

Pre-requisites

- Basic understanding of engineering or manufacturing
- Familiarity with product or process development
- No prior FMEA training required

Learning Objectives

- Understand the purpose, structure, and benefits of FMEA
- Differentiate between DFMEA and PFMEA
- Apply the AIAG-VDA 7-step methodology to assess risk
- Develop structured analysis using functional breakdown and failure chains
- Calculate Severity, Occurrence, and Detection ratings
- Use Action Prioritization (AP) to determine mitigation strategies
- Integrate FMEA with Process Flow and Control Plan

Course Agenda

1. Introduction to FMEA

- Definition, history, and evolution of FMEA
- Linkage with IATF 16949 and APQP
- When to perform DFMEA and PFMEA

2. AIAG-VDA FMEA Framework

- Key differences from traditional FMEA
- Introduction to the 7-Step process
- FMEA structure and scope definition

3. DFMEA – Design Failure Mode and Effects Analysis

- 7 Steps: Planning, Structure, Function, Failure, Risk, Optimization, Documentation
- Special characteristics and design controls
- Case study using a real-world part

4. PFMEA – Process Failure Mode and Effects Analysis

- Differences from DFMEA
- Mapping with process flow diagram
- Root cause identification and linkage with Control Plan

5. Action Prioritization and Optimization

- Eliminating RPN and using AP (High/Medium/Low)
- Targeting actions to improve detection/prevention
- Tracking and re-evaluating actions

6. FMEA Best Practices and Documentation

- Cross-functional team involvement
- Excel templates vs. software tools
- FMEA-MSR overview

What You'll Receive

- QLSS Certificate of Completion
- Training Material / Handout (PDF Format)
- Final Assessment with Individual Evaluation Report

Who Should Attend

- Product and Process Design Engineers
- Quality and Manufacturing Engineers
- Supplier Development and APQP Teams
- Internal Auditors and Documentation Owners
- Anyone involved in DFMEA or PFMEA preparation or review

Trainer Profile

Our trainers are experienced professionals with a strong background in product and process engineering, quality management, and automotive core tools. Each trainer has facilitated numerous FMEA workshops across industries and supported organizations during customer audits and third-party assessments. Their expertise lies in simplifying complex methodologies, driving cross-functional participation, and delivering result-oriented training aligned with industry expectations.

Instructional Methodology

- Clause-linked explanation to IATF 16949
- AIAG-VDA 7-Step templates walkthrough
- Real-life case studies for hands-on learning
- Group-based exercises with facilitator guidance
- Risk rating and decision-making practice

What Makes QLSS FMEA Training Unique

- Based on latest AIAG-VDA harmonized guidelines
- Clear guidance on transition from legacy to new FMEA
- Structured templates for PFMEA and DFMEA
- Activity-based practice to improve retention
- Audit-ready approach to FMEA documentation and ownership